

Hit Factors: What makes a song a hit?

1. Problem Statement

In the modern music industry, artists, producers, and labels are constantly trying to figure out the formula behind a successful song. The main objective of this project is to build a machine learning model that predicts whether a track will become a "hit" or not based strictly on its musical, textual, and structural characteristics, without relying on popularity metrics or stream counts.

Furthermore, the music landscape has drastically shifted over the decades due to the rise of streaming platforms (such as Spotify), social media (like TikTok), and changing consumption habits. To capture this evolution, our research will be split into two distinct eras:

- Era 1 (2000–2011): The pre-streaming/radio & digital downloads era.
- Era 2 (2012–2021): The streaming era (marked by shorter song structures, the boom of rap, playlist curation, etc.).

By training our models on both periods separately, we aim to uncover whether the ingredients of a hit song have fundamentally changed over time. Additionally, we will look into a secondary phase focusing exclusively on charting songs to analyze what drives a track to reach a higher peak position.

2. Input and Output Data & Schema

To avoid circular logic or data leakage, popularity, streaming counts, **and Genius views** are strictly excluded from our inputs, as they directly measure success rather than predict it.

Our project relies on two main target outputs:

- Part 1 (Classification) : `is_hit` = 1 if the song entered the Billboard Hot 100 chart, and 0 if it did not.
- Part 2 (Regression) : For charting songs only, `peak_position` represents their highest rank achieved on the Billboard chart.

To ensure smooth collaboration on GitHub, all input features and target variables are structured as follows:

Category	Variable Name	Type & Range	Description & Source
Output	<code>is_hit</code>	Binary (0 or 1)	Classification label (1 if Billboard Hot 100 entry).
Output	<code>peak_position</code>	Numerical	Highest rank achieved on Billboard (charting tracks only).
Audio Features	<code>duration_ms</code>	Numerical	Track length in milliseconds (Kaggle Spotify Dataset).

Audio Features	tempo	Numerical	Estimated tempo in beats per minute / BPM.
Audio Features	energy	Float (0.0 - 1.0)	Perceptual measure of intensity and activity.
Audio Features	danceability	Float (0.0 - 1.0)	Suitability of a track for dancing.
Audio Features	acousticness	Float (0.0 - 1.0)	Confidence measure of whether the track is acoustic.
Audio Features	valence	Float (0.0 - 1.0)	Musical positiveness conveyed by the track.
Audio Features	instrumentalness	Float (0.0 - 1.0)	Predicts whether a track contains no vocals.
Audio Features	speechiness	Float (0.0 - 1.0)	Detects the presence of spoken words.
Audio Features	explicit	Binary (0 or 1)	Flag indicating explicit content.
Lyrical Features	lyrical_sentiment	Categorical / Score	Polarity of lyrics (Genius Dataset).
Lyrical Features	vocab_richness	Numerical	Lexical diversity (unique vs. total words).
Lyrical Features	primary_theme	Categorical	Core subject of the lyrics (e.g., love, party).
Temporal Context	release_date	Date / Datetime	Exact release date via MusicBrainz API.
Genre Classification	genre	Categorical	Main musical genre classification.
Collaboration	is_feat	Binary (0 or 1)	Presence of featured artists (collaboration).
Business Context	label_type	Categorical	Record label type (major vs. indie).

3. Method, Approach

Our methodology follows a rigorous data science pipeline divided into several steps:

- Data Collection & Integration :
 - Merging the foundational Spotify Tracks Dataset with the Billboard Hot 100 tracking dataset.
 - Enriching the dataset with lyrical insights from Genius and precise metadata (release dates, labels) via the MusicBrainz API.
 - Centralizing our code, datasets, and resources on our collaborative GitHub repository: [Hit Factors GitHub](#).
- Data Preprocessing & Cleaning :
 - Handling missing values, normalizing numerical features, and encoding categorical variables.

- Splitting the consolidated dataset into the two targeted chronological eras (2000–2011 and 2012–2021) for fair comparisons.
- Model Training & Validation (Part 1 - Hit Prediction) :
 - Training classification models (e.g., Logistic Regression, Random Forest, or XGBoost) to predict the binary outcome (hit vs. non-hit).
 - Using cross-validation to evaluate model performance (F1-score, Precision, Recall, ROC-AUC).
- Feature Importance & Era Comparison :
 - Extracting feature importances to identify which factors weigh the most in predicting a hit.
 - Comparing weights between the 2000–2011 era and the 2012–2021 era to observe how industry drivers have evolved.
- Advanced Charting Analysis (Part 2) :
 - Filtering the dataset to include only charting tracks and applying regression models to understand what propels a song to peak higher.

4. Expected Results

Key Success Drivers Identification : We expect to identify the most critical audio and structural attributes that correlate with commercial success (e.g., optimal track duration, high danceability, or specific lyrical sentiments).

Era Evolution Insights : We anticipate discovering clear shifts in success patterns—for instance, expecting track duration to have a higher negative impact on hits in the 2012–2021 streaming era compared to the 2000–2011 era.

5. Project Resources & References

- Datasets & Code Repository : [Hit Factors GitHub Repository](#) *(All datasets, source files, and integration scripts are referenced and stored directly within this repository).*